

Local Fields of Characteristic 0

In these notes we will state a number of things that will lead to a description of the “local” behaviour of number fields. Most of the statements are given without detailed proofs as they involve techniques from topology and metric spaces which are not covered here.

Recall that a finite extension K of $\mathbb{Q}$ is called a *number field*. As seen earlier, there is an irreducible monic polynomial P in $\mathbb{Z}[T]$ such that K is generated by a root of P . In other words, K is isomorphic to $\mathbb{Q}[T]/\langle P(T) \rangle$.

Given any extension L of $\mathbb{Q}$, we have defined the L -algebra $L \otimes_{\mathbb{Q}} K$ which can be identified with $L[T]/\langle P(T) \rangle$. Since P is irreducible in $\mathbb{Q}[T]$, it has distinct factors in $L[T]$, we can write it as a product of distinct irreducible factors

$$P(T) = P_1(T) \cdots P_g(T) \text{ in } L[T]$$

Thus, if $L_i = L[T]/\langle P_i(T) \rangle$ for $i = 1, \dots, g$, then we have an isomorphism

$$L \otimes_{\mathbb{Q}} K \cong L_1 \times \cdots \times L_g$$

We are interested in the cases when $L = \mathbb{Q}_p$ is the field of p -adic numbers, or $L = \mathbb{R}$. In this case the fields L_i so obtained are *local fields of characteristic 0*. (This is a special case of the notion of locally compact fields. We will not explore this notion in full generality.)

Case $L = \mathbb{R}$

A polynomial P in $\mathbb{R}[T]$ splits as a product of linear and quadratic factors.

$$P(T) = (T - a_1) \cdots (T - a_r) \cdot (T^2 - b_1T + c_1) \cdots (T^2 - b_sT + c_s)$$

where $b_i^2 < 4c_i$ for $i = 1, \dots, s$.

Thus, in this case we have:

- $L_i = \mathbb{R}[T]/\langle T - a_i \rangle \cong \mathbb{R}$ for $i = 1, \dots, r$;
- $L_{r+i} = \mathbb{R}[T]/\langle T^2 - b_iT + c_i \rangle \cong \mathbb{C}$ for $i = 1, \dots, s$.

There are two choices of the isomorphism (which is identity on $\mathbb{R}$) in the second case.

General case

A finite field extension E of $\mathbb{Q}_p$ is a finite dimensional $\mathbb{Q}_p$ vector space. Hence, it has a natural topology. In fact, given a $\mathbb{Q}_p$ basis of E we can identify E with $\mathbb{Q}_p^n$ and define a norm $\|\cdot\|_0$ on E :

$$\|\alpha\|_0 = \|(a_1, \dots, a_k)\| = \max\{|a_i|_p : i = 1, \dots, k\}$$

The topology obtained using the metric $d(\alpha, \beta) = \|\alpha - \beta\|_0$ is independent of the choice of the basis.

The map $\alpha \mapsto \text{Nm}_{E/\mathbb{Q}_p}(\alpha)$ is a polynomial function of $(a_1, \dots, a_n)$ and is thus a *continuous* map from E to $\mathbb{Q}_p$ with respect to this topology.

Now, $\text{Nm}_{E/\mathbb{Q}_p}(\alpha) = 0$ if and only if $\alpha = 0$. So this gives a non-zero function for non-zero α .

Note that the set $\{\alpha : \|\alpha\|_0 = 1\}$ is a *compact* set in this topology. It follows that

$$0 < c \leq |\text{Nm}_{E/\mathbb{Q}_p}(\alpha)|_p \leq C \text{ whenever } \|\alpha\|_0 = 1$$

Note that $\text{Nm}_{E/\mathbb{Q}_p}(\alpha)$ is homogeneous of degree k for scaling by elements of $\mathbb{Q}_p$. It follows easily that

$$c\|\alpha\|_0^k \leq |\text{Nm}_{E/\mathbb{Q}_p}(\alpha)|_p \leq C\|\alpha\|_0^k \text{ for all } \alpha$$

Thus, we define

$$\|\alpha\|_E = |\text{Nm}_{E/\mathbb{Q}_p}(\alpha)|_p^{1/k}$$

to get an equivalent norm on E . Note that this norm is the same as $|\cdot|_p$ on elements of $\mathbb{Q}_p$.

The norm $\|\cdot\|_E$ on E extends the usual norm on $\mathbb{Q}_p$ such that the induced topology on E is that of a finite dimensional vector space over $\mathbb{Q}_p$.

We further note that the set

$$\mathcal{O}_E = \{\alpha : \|\alpha\|_E \leq 1\}$$

is a subring of E in which

$$m_E = \{\alpha : \|\alpha\|_E < 1\}$$

is the unique maximal ideal which is a principal ideal. We denote by π_E a generator of this maximal ideal which is unique upto a unit in $\mathcal{O}_E$. The field

$$F_E = \mathcal{O}_E/m_E$$

is a finite extension of $\mathbb{F}_p$. Hence, it is of the form $F_E \cong \mathbb{F}_p[T]/\langle Q(T) \rangle$ for some irreducible polynomial Q in $\mathbb{F}_p[T]$ of degree $f = [F_E : \mathbb{F}_p]$. We choose x to be an element of $\mathcal{O}_E$ whose image in F_E satisfies $Q(x) = 0$.

The F_E vector space m_E^r/m_E^{r+1} is a 1-dimensional vector spaces generated by the image of π_E^r . It follows that it is generated as a $\mathbb{F}_p$ vector space by $\{x^i \pi_E^r\}_{i=0}^{f-1}$.

Since p lies in m_E , we see that $p = \pi_E^e \cdot u$ where u is a unit in $\mathcal{O}_E$, for a suitable integer e .

One can show that the set $\{x^i \pi_E^j : i = 0, \dots, f-1; j = 0, \dots, e-1\}$ gives a basis for $\mathcal{O}_E$ over $\mathbb{Z}_p$.

Let us choose $\tilde{Q}$ in $\mathbb{Z}_p[T]$ which lifts the polynomial Q in $\mathbb{F}_p[T]$. We note that $\tilde{Q}(x)$ lies in m_E .

1. If $\|\tilde{Q}(x)\|_E = \|\pi_E\|_E$, then $\tilde{Q}(x) = \tilde{\pi}_E$ is an *alternate* choice for π_E . In this case, we see that $\mathcal{O}_E$ is generated by powers of x over $\mathbb{Z}_p$.
2. If $\|\tilde{Q}(x)\|_E < \|\pi_E\|_E$, we take $\tilde{x} = x + \pi_E$ and note that

$$\tilde{Q}(\tilde{x}) = \tilde{Q}(x) + \tilde{Q}'(x)\pi_E + \pi_E^2\beta$$

for some suitable β in $\mathcal{O}_E$. Since Q has distinct roots in F_E , one sees that $\tilde{Q}'(x)$ is a unit in $\mathcal{O}_E$. It follows that $\|\tilde{Q}(\tilde{x})\|_E = \|\pi_E\|_E$. Thus, one sees that the powers of $\tilde{x}$ generate $\mathcal{O}_E$ over $\mathbb{Z}_p$.

As a result, in both cases, we see that there is a polynomial P in $\mathbb{Z}_p$ such that $\mathcal{O}_E$ is isomorphic to $\mathbb{Z}_p[T]/\langle P(T) \rangle$.

A finite extension field of $\mathbb{Q}_p$ is the field of fractions of $\mathbb{Z}_p[T]/\langle P(T) \rangle$ where P is an irreducible monic polynomial in $\mathbb{Z}_p[T]$.